Impression materials

1-Complete

1) **………..is the time from mixing the impression material to the development of the elastic properties.**
2) **………..is the time from mixing the impression material to the complete development of the elastic properties.**
3) **Insertion of the impression material must be …….the end of working time.**
4) **Removal of the impression material must be …… the end of setting time.**
5) **Die is a ……. Reproduction.**
6) **The insertion of the impression material must be in a completely ……….. state.**
7) **The removal of the impression material must be a completely ……….. state.**
8) **The higher the flow, the ……the dimensional accuracy.**
9) **Alginate has …….. dimensional stability.**
10) **Hydrophilicity means …………….. contact angle between the impression material and mucosa.**
11) **Relation between upper and lower teeth (occlusion) is recorded by ………..**
12) **………………… are material with highly resistant to flexure, and it fractures suddenly under stress.**
13) **…………………. are flexible material and can be deformed and still return to its original form to a great extent when unstressed.**
14) **………………. is a heterogeneous mixture with dispersed phase having particle size intermediate between true solution and suspension.**
15) **The most flexible impression material is …………………………..**

16) **Colloid consists of two phases …………… and ……………**

17) **………………… is used as retarder in alginate as it reacts with…………………**

18) **………………….. is used to minimize dust in alginate.**

19) **................is used to control consistency of mixture.**

20) **Alginic acid is formed of ………… and ………………**

21) **Hydrogels could lose water if left in dry air by evaporation in a process called …… which result in………….**

22) **Hydrogels if submerged in water they will absorb water in a process called ……….. which result in………….**

23) **Increasing the amount of retarder in the alginate leads to increase the …………..**

24) **Increasing water temp during mixture result in …………..setting time.**

25) **The higher the water/powder ratio the………. tear strength, the …… the accuracy, and the ……… the setting time.**

26) **Setting mechanism of elastomers is ……………**

27) **The higher the filler content the……….. polymerization shrinkage, the ……….. the viscosity and the …………… the accuracy.**

28) **Simultaneous (dual viscosity) technique is also called………, where putty and light impression material has the same………..**

29) **The most flexible elastomer is…………………**

30) **The impression material which has the highest elastic recovery and dimensional stability is…………**

31) **The elastomer which has the lowest polymerization shrinkage is …………**

32) **Compared with elastomers, hydrocolloids has ………… tear strength.**

33) **The impression material which has the highest stiffness is ………..**

34) **Extra-low and putty forms are only for……….. and ……………..**

35) **The setting reaction mechanism of PE is……………**

36) Working and setting times of elastomeric impression materials are shortened by……………..

37) The sharp transition from plastic to elastic into the setting phase of PE is called ……….……..

38) ……………..are responsible for the pseudoplastic properties of PE.

39) …………………..or ….......... act as a retarder in Polysulfide.

40) The setting reaction of polysulfide is …………….. and the by product is………..

41) The most common catalyst in polysulfide is………………

42) Polysulfide needs ……..……………tray.

43) Pouring gypsum in polysulfide impression material mustn't exceed ……. minutes after removal.

44) The ….…..…. produced as a byproduct and causes high shrinkage in condensation silicon.

45) Impression compound is used with a …… tray.

46) Type I impression compound is ……. fusing material while type II is ……fusing material.

47) The setting reaction of ZnO/eugenol is……………..

48) The reaction of ZnO/eugenol occurs in presence of…………

49) It is contraindicated to disinfect ZnO/eugenol impression by …….…… but it can be disinfected by…….…..

2-MCQ

1. Which of the following impression materials would you use for taking secondary impression for a dentulous patient?

a.Addition silicone b. Alginate c. Impression compound d. Zinc oxide eugenol

2. Which of the following impression materials can be used ONLY for edentulous

patients?

a. Alginate b. Condensation silicone c. Polysulfide d.Zinc oxide eugenol

3. Which of the following materials has the highest elastic recovery?

a.Addition silicone b. Alginate c. Polysulfide d. Polyether

4. Which of the following materials is the most dimensionally stable?

a.Addition silicone b. Alginate. c. Condensation silicone d. Polyether

5. Which of the following materials set through a reversible physical reaction?

a.Agar

b. Alginate

c. Condensation sihcone

d. Zine oxide cugenol

6. Which of the following is an undesirable property of the impression material?

a.Dimensional stability

b.High viscosity at the time of insertion in the mouth

C Minimal dimensional changes during setting

d. Minimal permanent deformation upon removal from mouth

7- All the following impression materials are elastic, EXCEPT…….

a. zinc oxide eugenol b.polyether c.agar-agar d. addition silicone

8- Working time must be ……..than the time required for mixing, loading , and insertion of the tray

a. Equal b. greater c.less

9- time elapsed from the beginning of mixing, until complete setting or hardening occurs is known as:

a- Setting time b- working time c- mixing time d- initial setting time

10- Impression materials that have mechanical properties permitting considerable elastic deformation but that return into their original form are classified as:

a. elastic b. visous c. inelastic d. rigid

11-Which of the following dental impression materials is an example of an non-aqueous elastomers ?

a. impression compound

b. ZOE impression paste

c. Polysulfide

d. Irreversible hydrocolloid

12- Surfactant is added to :

A. Addition silicone.

B. Agar-Agar.

C. Plaster of Paris impression.

D. Alginate.

13- Which of the following materials is the stiffest?

A. Condensation silicon

B. Impression compound

C. Alginate

D. Polyether

14- Impression can be classified according to manner of setting into:

A. Inelastic and elastic impression

B. Physical and chemical set

C. Primary and secondary

D. Dentulous and edentulous

15- Flexibility of impression means:

A. Ease of removal from undercut
B. Recover to original shape when removed from undercut
C. Permanent deformed when removed from under cut
D. Fractured when removed from undercut

16- Loss of water by evaporation from the surface of hydrocolloids gel is called

A. Imbibition B. Memory C. Hysteresis D. Syneresis

17- Alginate impression material :

A. have higher tear strength

B. Is an irreversible hydrocolloid impression material

C. Used as a secondary impression

18- Alginate impression should be removed by sharp snap removal from the patient mouth to :

A. Increase permanent deformation

B. Increase tear strength

C. Decrease permanent deformation

D. B & C

19- The impression material with the highest tear strength is :

A. Addition silicon B. Polyether C. Polysulfide D. Agar hydrocolloid

20- The impression that is most easy to remove after setting is :

A. Addition silicon

B. Polyether

C. Polysulfide

D. Condensation silicon

21- Sulfur containing gloves should be avoided during mixing ……….. impression material as it affects its setting reaction:

A. Addition silicon

B. Polysulfide

C. Polyether

D. Impression compound

22- Which of the following impression water is essential for its setting:

A. Impression compound

B. Zinc oxide/eugenol

C. Polyether

D. None of the above

23. Material used for border molding is:

A. Addition silicon

B. polysulfide

C. polyether

D. Impression compound (green stick)

24- Alginate supplied in form of :

A. Two paste B. Powder and water liquid

25- Alginate impression set by:

A. Chemically B. Physically

26. Increase powder/liquid ratio in alginate:

A. Increased setting time (prolonged)

B. Not affect setting time

C. Decreased setting time (fast setting)

27- Heavy or putty consistency impression considers:

A. Secondary impression

B. Primary impression

C. Tray for light consistency impression

3-Short essay questions

1) **Is it preferred to use hydrophobic or a hydrophilic impression material and why?**
2) **Why is alginate only used to take a primary impression?**
3) **What would happen if we used an impression material with high viscosity?**
4) **Why can't we use impression compound with edentulous patients with undercut?**
5) **What is the difference between setting and working time?**
6) **Mention the ideal requirements of impression materials.**
7) **Give reason. Impression material must have high resistance to tear and high elastic recovery.**
8) **How does the impression material be compatible with model and die material?**
9) **Why does alginate have low dimensional stability?**
10) **How to overcome low tear strength of the alginate?**
11) **Why do we use a perforated tray with alginate?**
12) **Why is it essential to store alginate in 100% relative humidity or wrapping it with wet towel?**

13) **Why does putty elastomers have the least polymerization shrinkage?**
14) **How can we avoid the production of bubbles in gypsum dies or casts by the hydrogen gas when using addition silicon impression material?**
15) **What is the role of palladium in addition silicon impression material?**
16) **Why is it contraindicated to use latex gloves with addition silicon?**
17) **Which impression technique is preferred to be used with condensation silicone?**
18) **Why mustn't disinfection of PE exceed 10 minutes?**
19) **Why may PE cause skin irritation and how to avoid it?**
20) **Which disinfectant is forbidden to be used in disinfection of impression materials or any dental material?**
21) **Why do we use only a special tray with polysulfide?**
22) **How to accelerate the setting reaction of ZnO/eugenol?**
23) **Give reason. Addition silicon and polyether can be used in single-viscosity technique.**
24) **Give reason. You shouldn't leave the cast too long in contact with the alginate impression material after setting.**

II. Answer the following questions:

1. What impression material(s) would you use for making a primary impression and a secondary impression for the patient whose upper jaw is shown in the following figure?

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2. What is the type of this cast (primary or secondary)?
   What is the type of this tray? Which impression material would be used to make an impression with this tray?